

Project 1: Global Data Landscape

PPOL 5206: Massive Data Fundamentals

Spring 2026

Overview

The legal and physical environments for data vary significantly across the world and are constantly evolving. For this project, you will select a specific country to research and analyze its data ecosystem, examining both technical infrastructure and regulatory frameworks.

On the infrastructure side, you will investigate the presence and operations of major cloud providers, data center capacity and connectivity, internet access methods and penetration rates, and the country's readiness for advanced data-driven technologies such as AI, IoT, and cloud computing.

On the regulatory side, you will examine privacy and data protection laws, cross-border data transfer restrictions, cybersecurity requirements, and compliance frameworks that affect how organizations can collect, store, and process data.

Your analysis should consider how historical context, regional influences, economic conditions, and cultural factors shape the country's approach to data governance—and how these regulatory choices interact with infrastructure investment and technology adoption.

You are encouraged to include relevant comparisons with peer nations to provide context for your analysis.

Deliverables

This project culminates in two deliverables:

- A concise presentation (10–15 minutes) highlighting key findings, critical factors, and the future outlook for data systems in your chosen country.
- A written briefing report (less than 2 or 3 thousand words) providing deeper analysis of infrastructure, legal frameworks, and their interconnections.

Milestones

Milestone	Due Date	Description
Country Selection	January 29	Submit your country choice (first-come, first-served to avoid duplicates)

Outline & Sources	February 12	Submit a structured outline identifying key topics and at least 5 preliminary sources
Presentation	February 26	In-class presentation during our Data Ethics & Governance session
Final Report	March 11	Written briefing report submitted via Canvas

Guiding Questions

The following questions can help guide your research. This is not a required checklist—select the questions most relevant to your country and add others as appropriate.

Infrastructure & Connectivity

- What major cloud providers operate in the country? What data center presence do they have?
- What is the state of domestic data center infrastructure? Does the country rely primarily on foreign facilities?
- What are the predominant methods of internet access? What are broadband, fiber, and mobile penetration rates?
- How does submarine cable connectivity affect the country's international data flows?
- What is the digital divide between urban and rural areas? What socioeconomic factors affect access?

Legal & Regulatory Environment

- What privacy and data protection laws exist? What legal protections do citizens have regarding their data?
- How does the regulatory framework affect data storage, cross-border transfers, and data localization?
- What cybersecurity requirements apply to organizations operating in the country?
- How do national security concerns influence data policies and infrastructure investments?
- What international agreements or trade considerations influence data regulations?

Context & Future Outlook

- How do historical, regional, and cultural factors shape the country's data governance approach?
- How ready is the country for advanced data-driven technologies (AI, IoT, blockchain)?

- What are the environmental impacts of data infrastructure? How do sustainability initiatives shape decisions?
- What trends suggest where the country's data landscape is heading in the next 5–10 years?

Recommended Sources

The following sources provide a starting point for your research. You should seek additional sources specific to your country. Please cite all sources using APA format.

Global ICT & Digital Economy Reports

International Telecommunication Union (ITU) – Global ICT statistics, broadband penetration, and telecom infrastructure reports.

Website: <https://www.itu.int/en/ITU-D/Statistics/Pages/stat/default.aspx>

Citation: *International Telecommunication Union. (n.d.). Statistics.*
<https://www.itu.int/en/ITU-D/Statistics/Pages/stat/default.aspx>

World Bank – Digital Development – Internet access data, digital inclusion metrics, and economic impacts of digital policies.

Website: <https://www.worldbank.org/en/topic/digitaldevelopment>

Citation: *World Bank. (n.d.). Digital development.*
<https://www.worldbank.org/en/topic/digitaldevelopment>

UNCTAD Digital Economy Reports – Analysis of global digital policies, e-commerce readiness, and regulatory trends.

Website: <https://unctad.org/topic/e-commerce-and-digital-economy>

Citation: *United Nations Conference on Trade and Development. (n.d.). E-commerce and digital economy.* <https://unctad.org/topic/e-commerce-and-digital-economy>

OECD Digital Economy Outlook – Policy analysis on data infrastructure, internet governance, and digital transformation.

Website: <https://www.oecd.org/digital/>

Citation: *Organisation for Economic Co-operation and Development. (n.d.). Digital economy.* <https://www.oecd.org/digital/>

World Economic Forum – Global Competitiveness Reports – Digital competitiveness rankings and technology readiness indicators.

Website: <https://www.weforum.org/reports>

Citation: *World Economic Forum. (n.d.). Global competitiveness reports.*
<https://www.weforum.org/reports>

Portulans Institute – Network Readiness Index – Comprehensive index measuring countries' readiness to leverage ICT for development.

Website: <https://networkreadinessindex.org/>

Citation: Portulans Institute. (n.d.). Network Readiness Index.
<https://networkreadinessindex.org/>

Legal & Regulatory Frameworks

DLA Piper – Data Protection Laws of the World – Comparative privacy law tracker covering 100+ jurisdictions with regular updates.

Website: <https://www.dlapiperdataprotection.com/>

Citation: DLA Piper. (n.d.). Data protection laws of the world.
<https://www.dlapiperdataprotection.com/>

OneTrust DataGuidance – Detailed regulatory research platform with country-specific privacy law summaries.

Website: <https://www.dataguidance.com/>

Citation: OneTrust DataGuidance. (n.d.). Privacy research.
<https://www.dataguidance.com/>

Electronic Frontier Foundation – Global Surveillance – Research on government surveillance practices and privacy law enforcement.

Website: <https://www.eff.org/issues/global-surveillance>

Citation: Electronic Frontier Foundation. (n.d.). Global surveillance.
<https://www.eff.org/issues/global-surveillance>

Access Now – Digital Rights – Reports on international digital rights issues, internet shutdowns, and privacy advocacy.

Website: <https://www.accessnow.org/>

Citation: Access Now. (n.d.). Digital rights and privacy. <https://www.accessnow.org/>

Privacy International – Research on surveillance, data exploitation, and privacy rights worldwide.

Website: <https://privacyinternational.org/>

Citation: Privacy International. (n.d.). Research. <https://privacyinternational.org/>

Council of Europe – Convention 108+ – Legally binding international treaty on data protection with signatory information.

Website: <https://www.coe.int/en/web/data-protection/convention108>

Citation: Council of Europe. (n.d.). Convention 108+. <https://www.coe.int/en/web/data-protection/convention108>

European Commission – Adequacy Decisions – Official determinations of which countries meet EU data protection standards.

Website: https://commission.europa.eu/law/law-topic/data-protection/international-dimension-data-protection/adequacy-decisions_en

Citation: European Commission. (n.d.). Adequacy decisions.
<https://commission.europa.eu/law/law-topic/data-protection/>

APEC Cross-Border Privacy Rules – Asia-Pacific framework for cross-border data transfers and privacy certification.

Website: <https://www.apec.org/about-us/about-apec/fact-sheets/what-is-the-cross-border-privacy-rules-system>

Citation: Asia-Pacific Economic Cooperation. (n.d.). Cross-Border Privacy Rules System. <https://www.apec.org/>

Infrastructure & Connectivity Data

CIA World Factbook – Communications – Baseline statistics on internet penetration, telecom infrastructure, and connectivity.

Website: <https://www.cia.gov/the-world-factbook/>

Citation: Central Intelligence Agency. (n.d.). The world factbook. <https://www.cia.gov/the-world-factbook/>

Cloudflare Radar – Real-time internet traffic data, outage tracking, and connectivity trends by country.

Website: <https://radar.cloudflare.com/>

Citation: Cloudflare. (n.d.). Cloudflare Radar. <https://radar.cloudflare.com/>

Speedtest Global Index – Monthly rankings of mobile and fixed broadband speeds by country.

Website: <https://www.speedtest.net/global-index>

Citation: Ookla. (n.d.). Speedtest Global Index. <https://www.speedtest.net/global-index>

Akamai State of the Internet – Broadband adoption, connectivity quality, and cybersecurity threat reports.

Website: <https://www.akamai.com/resources/state-of-the-internet>

Citation: Akamai Technologies. (n.d.). State of the internet. <https://www.akamai.com/resources/state-of-the-internet>

CloudScene – Data Center Market Reports – Data center locations, cloud provider presence, and market analysis by region.

Website: <https://cloudscene.com/>

Citation: CloudScene. (n.d.). Data center reports. <https://cloudscene.com/>

Data Center Map – Interactive map of colocation data centers worldwide with facility details.

Website: <https://www.datacentermap.com/>

Citation: Data Center Map. (n.d.). Colocation data centers. <https://www.datacentermap.com/>

Submarine Cable Map – Interactive map of undersea internet cables showing landing points and capacity.

Website: <https://www.submarinecablemap.com/>

Citation: TeleGeography. (n.d.). Submarine Cable Map.
<https://www.submarinecablemap.com/>

GSMA Mobile Connectivity Index – Mobile network coverage, affordability, and digital services readiness scores.

Website: <https://www.gsma.com/r/mobile-connectivity-index/>

Citation: GSMA. (n.d.). Mobile Connectivity Index. <https://www.gsma.com/r/mobile-connectivity-index/>

Internet Society – Global Internet Reports – Analysis of internet governance, infrastructure development, and digital inclusion.

Website: <https://www.internetsociety.org/>

Citation: Internet Society. (n.d.). Global Internet Reports.
<https://www.internetsociety.org/>

Freedom House – Freedom on the Net – Annual assessment of internet freedom, censorship, and digital rights by country.

Website: <https://freedomhouse.org/report/freedom-net>

Citation: Freedom House. (n.d.). Freedom on the Net.
<https://freedomhouse.org/report/freedom-net>

Evaluation Criteria

Presentation (40 points)

Category	Points	Description
Content	15	Clear explanation of key findings; well-researched analysis covering both infrastructure and legal dimensions
Organization	10	Logical flow; structured slides that guide the audience through your analysis
Visual Aids	5	Effective use of maps, charts, and data visualizations to support your narrative
Delivery	5	Clear speaking; professional demeanor; appropriate pacing and audience engagement
Q&A	5	Ability to answer questions confidently and demonstrate depth of knowledge

Written Report (60 points)

Category	Points	Description
Research & Analysis	20	Depth of analysis; use of credible sources; insightful connections between infrastructure and regulation
Organization & Clarity	10	Logical structure; clear writing; well-organized sections with smooth transitions

Use of Data & Visuals	10	Effective integration of statistics, charts, tables, and maps to support arguments
Legal Analysis	10	Thorough explanation of regulatory frameworks; understanding of compliance requirements and their implications
Writing Mechanics	5	Grammar, spelling, and proper APA citations throughout
Conclusions	5	Synthesis of key takeaways; meaningful comparisons; actionable insights or recommendations

Collaboration

You may work individually or in teams of up to three students. Teams should ensure equitable distribution of work, and all team members will receive the same grade. If working in a team, please indicate team membership when submitting your country selection.

Good luck! I look forward to seeing your projects.